

Alejandro Rodríguez del Bosque

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Education

Purdue University – Graduation Date: May 2028

Research: Learning-integrated control architectures in embodied autonomous systems, emphasizing on autonomous decisionmaking.

Relevant Coursework: Artificial Intelligence, Introduction to Robotic Systems

Tecnológico de Monterrey, Campus Monterrey – Graduation Date: June 2026

B.S. in Robotics and Digital Systems Engineering – 3.7 GPA

Relevant Coursework: Control Systems; System & Signal Analysis; Advanced Embedded Systems; Data Structures & Algorithms; Intelligent Robotics Implementation

Experience

Human Centered Robotics Laboratory, The University of Texas at Austin

August 2025 - December 2025

Undergraduate Research Intern

- Researched learning-based control strategies for embodied robotic systems, focusing on stability and adaptive behavior in dynamic environments.
- Designed and implemented reinforcement learning policies using IsaacLab and PyTorch, achieving successful quadruped walking gaits under external disturbances.
- Developed and validated high-fidelity IsaacSim environments for locomotion and navigation benchmarking, and analyzed multi-sensor data to validate a drivetrain-based contact measurement framework

IEEE-RAS, Tecnológico de Monterrey

August 2025 - June 2026

Co-Founder & Vice President

- Co-founded the IEEE-RAS student branch and led the ANYmal-D research initiative, directing work on learning-based control, robust perception, and dynamic-terrain navigation.
- Coordinated multidisciplinary robotics projects integrating ROS 2, Gazebo, and IsaacSim, developing simulation-to-real validation workflows for autonomous systems.
- Designed and delivered reinforcement learning workshops and robotics research training, covering RL implementation, experimental design, reproducibility, and control-system validation for the 2025 Latin American Robotics Symposium (LARS).

Purdue University

August 2026 - Present

Graduate Teaching Assistant

- Led hands-on laboratory sessions, mentoring students through CAD, product data management, and engineering change workflows while troubleshooting technical challenges and coordinating course activities.
 - Managed student project evaluations and supported Teamcenter-based design release processes, reinforcing engineering documentation, collaboration, version control, and industry-standard product lifecycle practices.
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Selected Technical Projects

Puzzlebot Challenge for Implementation of Intelligent Systems Class

March – June 2025

Developer

- Designed the system architecture and ROS 2 communication framework, coordinating nodes and topics across an NVIDIA Jetson Nano, LiDAR, FPGA, encoder, and perception/control modules.
 - Developed closed-loop PID control and weighted A obstacle-avoidance algorithms, integrating LiDAR-based SLAM for autonomous navigation while maintaining safe distances from obstacles.
 - Developed the perception and task-execution pipeline using YOLO for company-logo detection and QR-code recognition, enabling autonomous package identification and delivery to designated storage locations.
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Skills

Programming: Python, C++, C, C#, Java, Assembly

Robotics & Simulation: ROS 2, IsaacSim, IsaacLab, Gazebo, Linux/Ubuntu, Git

Control & Autonomy: PID, State Feedback, Nonlinear Control, Kinematic Modeling, State Estimation, Motion Planning, SLAM

Machine Learning & Perception: PyTorch, Reinforcement Learning, YOLOv8, scikit-learn, Open3D, Computer Vision

Embedded & Systems: NVIDIA Jetson, FPGA, Sensor Integration, Multi-Sensor Data Analysis

Languages: Spanish (Native), English (Native-level), German (Basic)

